

(a)	Describe the fundamental characteristics of living organisms	1	7	3
(b)	Describe the functions of cell membrane, cell wall and cytoplasm	1	7	2
(c)	What are the differences between the plant cells and animal cells?	1	7	2
Q3.	Answer any two:			
(a)	What are enzymes? Discuss their structure, characteristics, and significance as biocatalysts in biological processes.	2	7	2
(b)	Define pesticides. Write down four advantages and four disadvantages of pesticides.	2	7	2
(c)	Define Fermentation. Discuss the different steps of Fermentation	2	7	3
Q4.	Answer any two:			
(a)	Elaborate how energy flow in the ecosystem is governed by the two laws of thermodynamics.	3	7	2
(b)	Discuss in details, about the functions of the ecosystem.	3	7	3
(c)	What are tropic levels? Discuss each tropic level with example.	3	7	2
Q5.	Answer the following:			
(a)	What is acid rain? Discuss with reactions, the harmful effects of acid rain.	4	7	3
(b)	Write a short note on noise pollution.	4	7	2
Q6.	Answer the following:			
(a)	What is e-wastes? Cite their common sources and composition. Discuss the steps involved in e-waste recycling.	5	7	3
(b)	What are the three phases of waste water treatment? Discuss with diagrams, the working principles of trickling filter.	5	7	2

Best of Luck

06/08/25 - 2nd sitting

G. V. Ramani Global University
Odisha

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Registration No. :

Total number of printed pages – 02

B.Tech
Subject Code: CY202

3rd/4th SEMESTER SUMMER QUARTER EXAMINATION – 2025
SUBJECT NAME: BIOLOGY AND ENVIRONMENTAL SCIENCE

BRANCH: B. Tech ALL BRANCHES

Time – 3 Hours

Full Marks – 100

Answer all Questions from Part – A and Part – B

Q1.	Part-A (ANSWER ALL QUESTIONS)	CO	Marks	BL
(a)	Draw a labelled diagram of mitochondria.	1	2	1
(b)	Write down four important characteristics of lipids.	1	2	2
(c)	What are proteins? Give examples of two protein rich food items	1	2	2
(d)	Write the full form of HIV and AIDS.	2	2	1
(e)	Write down any two advantages of Bioenergy.	2	2	2
(f)	Write down any two importance of medicinal plants in modern healthcare.	2	2	1
(g)	State the laws of thermodynamics followed in energy flow in ecosystem	3	2	2
(h)	What is food chain? Explain in brief with schematic diagram	3	2	3
(i)	What do you mean by nitrogen fixation?	3	2	2
(j)	Define particulate matter.	4	2	2
(k)	Write full forms: NAAQS, EPA, VOC, CFC, CCS etc.	4	2	1
(l)	What is carbon sequestration?	4	2	3
(m)	What are the three types of water quality parameters. Mention with examples.	5	2	2
(n)	What is activated sludge?	5	2	3
(o)	Mention any two goals of circular economy.	5	2	3

Part-B

CO Marks BL

Q2. Answer any two:

Part-B		CO	Marks	BL																		
Q2. Answer any two:																						
(a)	Compare and contrast prokaryotic and eukaryotic cells	1	7	3																		
(b)	Elaborate the structure and functions of mitochondria.	1	7	2																		
(c)	Differentiate between rough endoplasmic reticulum and smooth endoplasmic reticulum, in details, with diagram.	1	7	3																		
Q3. Answer any two:																						
(a)	What are biofertilizers? How are biofertilizers applied? Write down the two advantages and two challenges of biofertilizers.	2	7	1,2																		
(b)	Discuss competitive, non-competitive and un-competitive enzyme inhibition.	2	7	4																		
(c)	What are pesticides? Write down four advantages and four disadvantages of pesticides.	2	7	1																		
Q4. Answer any two:																						
(a)	Describe the processes involved in nitrogen cycle	3	7	3																		
(b)	How does energy flow in ecosystem. Discuss elaborately.	3	7	3																		
(c)	Discuss environmental assessment impact.	3	7	3																		
Q5. Answer the followings:																						
(a)	Discuss the characteristics, primary source, and harmful effects of the criterion air pollutants.	4	7	3																		
(b)	What is ozone layer depletion? How does CFC affect the ozone depletion?	4	7	2																		
Q6. Answer the followings:																						
(a)	What is e-wastes? Cite their common sources and composition. Discuss the steps involved in e-waste recycling.	5	7	2																		
(b)	The results from a BOD test, diluted by 100, are given in the Table. Calculate BOD ₃ , BOD ₅ and BOD ₇	5	7	4																		
<table border="1"> <thead> <tr> <th>Time (days)</th> <th>Diluted sample DO (mg/L)</th> <th>BOD (calculated) (mg/L)</th> </tr> </thead> <tbody> <tr> <td>0</td> <td>7.95</td> <td>-----</td> </tr> <tr> <td>1</td> <td>3.75</td> <td>420</td> </tr> <tr> <td>3</td> <td>2.75</td> <td>520</td> </tr> <tr> <td>5</td> <td>1.80</td> <td>615</td> </tr> <tr> <td>7</td> <td>0.78</td> <td>710</td> </tr> </tbody> </table>		Time (days)	Diluted sample DO (mg/L)	BOD (calculated) (mg/L)	0	7.95	-----	1	3.75	420	3	2.75	520	5	1.80	615	7	0.78	710			
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Registration No. :

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Total number of printed pages – 02

B.Tech
Subject Code: CY202

4th SEMESTER REGULAR EXAMINATION – 2025
SUBJECT NAME: BIOLOGY AND ENVIRONMENTAL SCIENCE
BRANCH: CSE (Group - 1 to 5), ECE, RAI, Civil
Time – 3 Hours
Full Marks – 100
Answer all Questions from Part – A and Part – B

Q1.	Part-A (ANSWER ALL QUESTIONS)	CO	Marks	BL
(a)	Draw a labelled diagram of a bacterial cell.	1	2	1
(b)	Mention two examples disaccharide with their constituent monosaccharides.	1	2	1
(c)	Write down two important functions of mitochondria.	1	2	1
(d)	Name the disease that appears due to the deficiency of Vitamin A and Iodine.	2	2	3
(e)	What are the different types of enzyme inhibition?	2	2	2
(f)	Write the chemical skeletal structure of steroids.	2	2	1
(g)	What do you mean by weathering in Phosphorus cycle?	3	2	2
(h)	Write down four importance of nitrogen cycle.	3	2	3
(i)	A producer yields a total productivity of 9×10^4 Kcal/m ² /year. It suffers a loss of productivity due to internal respiration of 3×10^3 Kcal/m ² /year. Compute the net productivity of the producer.	3	2	2
(j)	Write the full forms of EPA and VOC.	4	2	2
(k)	Define noise pollution.	4	2	2
(l)	Calculate the sound pressure level in decibels for a given sound pressure of 60×10^4 Micro-Pascal. <i>100 dB</i>	4	2	3
(m)	What is the saturation concentration of Dissolved oxygen (DO) at 20°C and 0°C?	5	2	2
(n)	What is activated sludge?	5	2	2
(o)	Write any four harmful effects of e-waste.	5	2	3

Part-B		CO	Marks	BL												
Q2. Answer any two:																
(a)	Elaborate on different types of organic constituents present in the living cells with examples.	1	7	3												
(b)	What are the differences between the plant cells and animal cells?	1	7	2												
(c)	Mention the physical significance of primary structure of proteins. Depict the primary structure of the following peptide code: FYP	1	7	2												
Q3. Answer any two:																
(a)	Define bioenergy. Mention its different forms and limitations. Enlight upon the common production processes of bioenergy.	2	7	2												
(b)	What are pesticides? Write down four advantages and four disadvantages of pesticides.	2	7	2												
(c)	What are vaccines? Name three commonly used vaccines. Explain how do vaccines work in our body.	2	7	3												
Q4. Answer any two:																
(a)	What are limiting factors in an environment? Explain a tolerance curve for a single species population existing on a single environmental gradient.	3	7	2												
(b)	Discuss with a suitable diagram, the different steps involved in the nitrogen cycle operating in nature.	3	7	3												
(c)	Consider a thermoelectric power plant of 915 MW of load factor of 72.5% and an efficiency of 40%. Determine the amount produced of particulates if coal is used. Given that the calorific value of coal is 29.7 MJ/kg; 80% of ash is fly ash (particulate).	4	7	4												
Q5. Answer the following:																
(a)	Compute the hydrocarbon discharge from a population centre of 1 million, if, (i) 300,000 vehicles travel 12,000 km per annum emitting 1 g/kg each (ii) The per capita consumption of oil-based paint is 2 litres per annum with a hydrocarbon content of 1 kg/l (iii) The per capita hydrocarbon from dry-cleaning solvents is 1 kg per annum	4	7	4												
(b)	What is greenhouse effect? How does greenhouse work? Give the examples of four major greenhouse gases. Discuss the adverse effect of greenhouse effect.	4	7	2												
Q6. Answer the following:																
(a)	The results from a BOD test, diluted by 100, are given in the Table. Calculate BOD ₃ , BOD ₅ and BOD ₇	5	7	4												
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(b)	What is integrated solid waste management (ISWM)? Explain elaborately the four functional elements of ISWM	5	7													



06/04/26 - 2nd sitting.

**C. V. Raman Global University
Odisha**

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Registration No. :

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Total number of printed pages – 02

B.Tech

Subject Code: CY202

END SEMESTER REGULAR EXAMINATION – 2026
SUBJECT NAME: BIOLOGY AND ENVIRONMENTAL SCIENCE
BRANCH: CSE, AIDS, DS, ECE, EE, EEE, RAI

Time – 3 Hours

Full Marks – 100

Answer all Questions from Part – A and Part – B

Q1.	Part-A (ANSWER ALL QUESTIONS)	CO	Marks	BL
(a)	What do you mean by a glycosidic bond?	1	2	1
(b)	Mention two aromatic amino acids.	1	2	1
(c)	What are ribosomes and cite its functions.	1	2	1
(d)	Name the disease that appears due to the deficiency of Calcium and Iodine.	2	2	1
(e)	What are corticosteroids. Give two examples of it.	2	2	2
(f)	Cite any two uses of each, Brahmi and Ashwagandha.	2	2	1
(g)	What are structural components of an ecosystem.	3	2	1
(h)	A producer yields a total productivity of 9×10^4 Kcal/m ² /year. It suffers a loss of productivity due to internal respiration of 5×10^3 Kcal/m ² /year. Compute the net productivity of the producer.	3	2	4
(i)	What do you mean by weathering in Phosphorus cycle?	3	2	2
(j)	Compute the amount of carbon dioxide (CO ₂) is emitted during the combustion of 16gm of methane (CH ₄)?	4	2	4
(k)	Mention the components of photochemical smog.	4	2	2
(l)	Determine the sound pressure level if the rms pressure value is given as 200 μ Pa.	4	2	3
(m)	The carbonate and non-carbonate hardness of the water sample are 300 ppm and 270 ppm respectively. Calculate the amount of permanent hardness.	5	2	4
(n)	Calculate the pH of 10^{-3} M HCl and 10^{-3} N HCl solutions.	5	2	4
(o)	What is the significance of 3R in waste management.	5	2	1

Part-B		CO	Marks	BL												
Q2. Answer any two:																
(a)	Describe the fundamental characteristics of living organisms.	1	7	3												
(b)	Differentiate between the animal cells and bacterial cells with detailed diagram.	1	7	2												
(c)	Write a note on the different constituent macromolecules in the human body. Mention their functions.	1	7	2												
Q3. Answer any two:																
(a)	What are Steroids? Discuss the functions and significance of steroids with examples.	2	7	2												
(b)	Define biofertilizers. Discuss the three types of biofertilizers with examples.	2	7	2												
(c)	What are vaccines? Discuss the working principle of any three broad category of vaccines.	2	7	3												
Q4. Answer any two:																
(a)	Elaborate on each trophic level with appropriate example.	3	7	2												
(b)	Discuss with diagram, the different stages of nitrogen cycle.	3	7	3												
(c)	What do you understand by sustainability? Discuss about the different components of any one ecosystem.	3	7	2												
Q5. Answer the following:																
(a)	What is greenhouse effect? How do greenhouse work? Give the examples of four major greenhouse gases. Discuss the adverse effect of greenhouse effect.	4	7	3												
(b)	What is acid rain? Discuss with reactions, the harmful effects of acid rain.	4	7	2												
Q6. Answer the following:																
(a)	The results from a BOD test, diluted by 100, are given in the Table. Calculate BOD ₃ , BOD ₅ and BOD ₇ <table><tr><th>Time (days)</th><th>Diluted sample DO (mg/L)</th></tr><tr><td>0</td><td>7.95</td></tr><tr><td>1</td><td>3.75</td></tr><tr><td>3</td><td>2.75</td></tr><tr><td>5</td><td>1.80</td></tr><tr><td>7</td><td>0.75</td></tr></table>	Time (days)	Diluted sample DO (mg/L)	0	7.95	1	3.75	3	2.75	5	1.80	7	0.75	5	7	3
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(b)	What is e-waste? Cite their common sources and composition. Discuss the steps involved in e-waste recycling.	5	7	3												

Best of Luck



Registration No. :

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Total number of printed pages – 02

B.Tech
Subject Code: CY202

END SEMESTER REGULAR EXAMINATION – 2025
SUBJECT NAME: BIOLOGY AND ENVIRONMENTAL SCIENCE

BRANCH: CSE (Group -6 to 10)/Civil/CH/ME/MR

Time – 3 Hours

Full Marks – 100

Answer all Questions from Part – A and Part – B

Q1.	Part-A (ANSWER ALL QUESTIONS)	CO	Marks	BL
(a)	Draw a labelled diagram of a prokaryotic cell.	1	2	1
(b)	Mention one function of endoplasmic reticulum.	1	2	1
(c)	Differentiate plant cell and animal cell.	1	2	1
(d)	Define bioenergy.	2	2	3
(e)	Mention the three types of enzyme inhibition.	2	2	2
(f)	Cite any two uses of Neem.	2	2	1
(g)	What do you mean by assimilation?	3	2	2
(h)	A producer yields a total productivity of 6×10^4 Kcal/m ² /year. It suffers a loss of productivity due to internal respiration of 4×10^3 Kcal/m ² /year. Compute the net productivity of the producer.	3	2	3
(i)	Mention one examples of autotrophs and primary herbivores.	3	2	2
(j)	Name the major six air pollutants.	4	2	2
(k)	What is smog?	4	2	2
(l)	Determine the sound pressure level if the rms pressure value is given as 60 μ Pa.	4	2	4
(m)	Mention one physical and one chemical water quality parameter.	5	2	2
(n)	What is eutrophication?	5	2	3
(o)	Mention any two goals of circular economy.	5	2	3